

DeltaGrad Optimization: Finite vs. Infinite Horizon (EMA) and Benchmark Protocols

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1 Introduction

Modern adaptive optimizers like Adam often oversmooth gradients, leading to noise memorization and instability. DeltaGrad introduces a dynamic Reliability Metric (R_t) based on instantaneous gradient coherence to modulate parameter update steps, significantly reducing gradient variance and improving generalization under label noise and high learning rates.

2 Non-Infinite Horizon (Finite Window K)

$$R_t = \text{clamp} \left(\frac{K - \sum_{j=0}^{K-1} \alpha^j \Phi_j}{K}, A, B \right) \quad g_s^{(t)} = \sigma g_s^{(t-1)} + (1 - \sigma) g^{(t)}$$
$$\Phi_j = \frac{|\tilde{g}^{(t)} - \tilde{g}^{(t-j)}|}{|\tilde{g}^{(t)}| + |\tilde{g}^{(t-j)}| + \epsilon} \quad \theta_{t+1} = \theta_t - \mu \cdot (R_t \odot m_t)$$

3 Infinite Horizon (EMA Formulations)

3.1 Base State Tracking Equations

$$\phi_t = \frac{|\tilde{g}^{(t)} - \tilde{g}^{(t-1)}|}{|\tilde{g}^{(t)}| + |\tilde{g}^{(t-1)}| + \epsilon} \quad m_t = \beta_m m_{t-1} + (1 - \beta_m) g^{(t)}$$
$$S_t = \beta_\phi S_{t-1} + (1 - \beta_\phi) \phi_t \quad \theta_{t+1} = \theta_t - \mu \cdot (R_t \odot m_t)$$
$$\hat{S}_t = \frac{S_t}{1 - \beta_\phi^t} \quad (\text{Bias Corrected})$$

3.2 Reliability Metric (R_t) Transformation Options

Below are candidate functions to transform accumulated disagreement $\hat{S}_t$ into $R_t \in [A, B]$:

Option 0: Baseline Linear Subtraction

$$R_t = \text{clamp} \left(1 - \hat{S}_t, A, B \right)$$

Option 1: Exponential Sensitivity Decay (Recommended)

$$R_t = \text{clamp} \left(\exp(-\gamma \cdot \hat{S}_t), A, B \right) \quad \text{where } \gamma > 0 \text{ controls gain}$$

Option 2: Inverse Soft Scaling

$$R_t = \text{clamp} \left(\frac{1}{1 + \gamma \cdot \hat{S}_t}, A, B \right) \quad \text{where } \gamma > 0$$

Option 3: Power / Root Gain Transformation

$$R_t = \text{clamp} \left(1 - (\hat{S}_t)^p, A, B \right) \quad \text{where } p \in (0, 1) \text{ (e.g., } p = 0.5)$$

Option 4: Sigmoidal / Threshold Damping

$$R_t = \text{clamp} \left(\sigma \left(\frac{\tau - \hat{S}_t}{s} \right), A, B \right) \quad \text{where } \tau = \text{threshold, } s = \text{scale}$$

Option 5: Self-Calibrating Z-Score Standardization

$$R_t = \text{clamp} \left(1 - \frac{\hat{S}_t}{\mu_S + k \cdot \sigma_S + \epsilon}, A, B \right)$$

4 Proposed Experimental Benchmarks

To systematically evaluate DeltaGrad-EMA against DeltaGrad (Windowed) and baseline optimizers (Adam, AdamW, SGD-Momentum, AdaGrad, RMSProp), experiments are divided into baseline replication setups and modern extended benchmarks.

4.1 Replication Benchmarks (Original DeltaGrad & Adam Setups)

1. CIFAR-100 5-Layer CNN (DeltaGrad Abstract Setup):

- *Architecture*: 3 Convolutional layers followed by 2 Fully Connected layers.
- *Data Noise Test*: Train under 0% clean data vs. 20% synthetic label noise (randomly flipped labels) with batch size 512 over 50 epochs.
- *Learning Rate Stress Test*: Set base LR via Optuna tuning (batch size 16), then scale by $1\times$, $3\times$, and $10\times$ across 5 independent runs to measure validation accuracy standard deviation and R_t -variance correlation.
- *Gradient Variance Tracking*: Log global mean gradient variance per batch to verify an order-of-magnitude reduction relative to Adam.

2. Convex & Sparse Objectives (Adam Paper Setup):

- *MNIST Logistic Regression*: L2-regularized convex log-likelihood with $\alpha_t = \alpha/\sqrt{t}$ decay and batch size 128.
- *IMDB Movie Reviews Bag-of-Words*: 10,000-dimensional sparse feature classification with 50% dropout noise (batch size 128) to test optimization under highly sparse gradients.

3. Non-Convex Vision & Generative Baselines (Adam Paper Setup):

- *MNIST MLP*: 2 fully connected hidden layers (1,000 ReLU units each) evaluated under deterministic loss and dropout noise (batch size 128).
- *CIFAR-10 ConvNet*: 3 alternating stages of 5×5 convolutions and 3×3 max-pooling followed by a 1,000-unit FC layer (batch size 128) to test cross-layer parameter adaptation.
- *MNIST VAE*: Variational Autoencoder (500 softplus units, 50-dimensional Gaussian latent space) to test stability against early training divergence.

4.2 Extended Benchmarks (Modern Vision, NLP, & Efficiency)

1. Modern Vision Backbones (ResNet-18 / ResNet-50):

- Evaluate convergence, final test accuracy, and label noise resilience on CIFAR-100 and ImageNet-1K using standard residual architectures.

2. Language Modeling / Transformers (NanoGPT on WikiText-2 / TinyStories):

- Train a decoder-only Transformer (6 to 12 layers) to verify that DeltaGrad-EMA scales to high-dimensional parameter spaces where windowed gradient state storage ($K \cdot d$) is memory-prohibitive.

3. Ablation & Overhead Profiling:

- *State Memory Footprint*: Compare state overhead ($3d$ for DeltaGrad-EMA vs. $2d$ for Adam vs. $(K + 1)d$ for DeltaGrad-Windowed).
- *Wall-Clock Time*: Measure per-epoch runtime overhead across batch sizes (16, 128, 512) to confirm overhead remains below 0.5%.
- *Decay Sensitivity* (β_ϕ): Sweep $\beta_\phi \in \{0.70, 0.80, 0.90, 0.95\}$ to determine the optimal effective lookback memory $K_{\text{eff}} \approx \frac{1}{1-\beta_\phi}$.

4.3 Summary of Experimental Configurations

Task / Dataset	Model Architecture	Batch Size	Epochs	Primary Metric / Focus
CIFAR-100 (Clean / 20% Noise)	5-Layer CNN	16 / 512	50	Noise memorization vs. generalization
CIFAR-100 (LR Stress Test)	5-Layer CNN	16	50	Acc. Std Dev across $1\times, 3\times, 10\times$ LR
MNIST	Multi-Class LogReg	128	45	Convex baseline convergence
IMDB BoW (10,000-dim)	LogReg + 50% Dropout	128	175	Sparse gradient optimization
MNIST	2-Layer MLP (1000 ReLU)	128	200	Non-convex optimization [8]
CIFAR-10	3-Stage ConvNet	128	45	Multi-layer scale adaptation
MNIST	VAE (500 units, 50 latent)	128	100	Initialization bias stability [1]
CIFAR-100 / ImageNet	ResNet-18 / ResNet-50	128	100	Modern deep backbone scaling
WikiText-2 / TinyStories	NanoGPT (Decoder-only)	32 / 64	20	High-dim parameter efficiency

Table 1: Comprehensive experimental suite combining original DeltaGrad and Adam paper setups with extended modern benchmarks.